BY ORDER OF THE COMMANDER
TRAVIS AIR FORCE BASE (AMC)

TRAVIS AIR FORCE BASE
INSTRUCTION 15-101

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Weather

WEATHER SUPPORT

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This instruction implements AFI 10-206, *Operational Reporting*, AFI 10-2501, *Air Force Emergency Management (EM) Program*, AFI 11-208 IP, *Department of Defense Notice to Airmen (NOTAM) System*, AFI 15-128, *Air Force Weather Roles and Responsibilities*, AFMAN 10-2504, *Air Force Incident Management Guidance for Major Accidents and Natural Disasters*, AFMAN 15-111, *Surface Weather Observations*, AFMAN 15-124, *Meteorological Codes*, AFMAN 15-129, *Air and Space Weather Operations*, and Air Mobility Command Instruction (AMCI) 15-101, *Weather Operations and Support*. It establishes responsibilities, weather support procedures and provides general information for weather services, including weather observations and forecasts, weather warnings, watches, and advisories (WWA); space weather data, information dissemination, and base-wide reciprocal support. It applies to units assigned to the 60th Air Mobility Wing (60 AMW) and units assigned, attached, or supported by Travis Air Force Base (AFB). This instruction applies to the 349th Air Mobility Wing (349 AMW) (AFRC), the 621st Contingency Response Wing (621 CRW), and the Navy VQ-3 Detachment. Ensure all records created as a result of processes prescribed in this publication are maintained in accordance with (IAW) AFMAN 33-363, *Management of Records*, and disposed of IAW the Air Force Records Information System (AFRIMS) Records Disposition Schedule (RDS) located at **https://www.my.af.mil/gcss-af61a/afrims/afrims/**. Refer recommended changes and questions about this publication to the Office of Primary Responsibility (OPR) using the AF Form 847, *Recommendation for Change of Publication*; route AF Forms 847 from the field through the appropriate functional's chain of command.

SUMMARY OF CHANGES

A special review of this instruction was conducted to capture procedural changes in AFMAN 15-129, *Air and Space Weather Operations.* This document includes revisions that pertain to the Weather Flight's assumption of local forecast, local mission support and Weather Watch responsibility in **AFMAN 15-129 para. 4.7, 6.2.2, and 7.3.2.**

Chapter 1

WEATHER FLIGHT (WF), TANKER AIRLIFT CONTROL CENTER (TACC) & OPERATIONAL WEATHER SQUADRON (OWS) INTERACTIONS

1.1. General. The 25th Operational Weather Squadron (25 OWS), the 618th Air Operations Center (Tanker Airlift Control Center) Weather Directorate, (618 AOC)/XOW), and the 60th Operations Support Squadron Weather Flight (60 OSS/OSW) are the official weather information agencies for Travis AFB, California. These agencies provide weather information in support of 60 AMW, 349 AMW, 621 CRW, Navy VQ-3 Detachment, and subordinate units and units assigned, attached, or supported by Travis AFB. The 618 AOC/XOW is commonly referred to as TACC/XOW throughout this document. The 60 OSS/OSW is commonly referred to as the Weather Flight (WF) and is the focal point for all weather-related issues. This instruction is reviewed and revised no greater than biennially or IAW with host/parent unit procedures if the time is less than biennially.

1.2. Concept of Operations. The 25 OWS at Davis Monthan AFB, Arizona, provides theater support, aviation services, and overwatch functions supporting WF/Dets in the primary area of interest.

1.2.1. The WF and TACC/XOW are the primary source of tailored weather services in support of the 60 AMW, the 349 AMW, the 621 CRW, various headquarters elements, and visiting aircrews. The WF and TACC/XOW make every effort to ensure that mission-limiting weather is anticipated and exploited, and that safety and Resource Protection (RP) are maintained.

1.3. Responsibilities. General responsibilities of the OWS and WF are outlined in **AFI 15-128**, **para. 2.3.3** and **para. 2.3.4** respectively.

1.3.1. The 25 OWS issues Special Weather Statements (SWS), and may provide flight weather briefings to transient aircrews passing through Travis AFB. Additionally, the OWS issues required WWAs in the event that the airfield is closed or the WF is unable, when coordinated with a Support Assistance Request (SAR).

1.3.2. The WF issues all Watches, Warnings and Advisories (WWA's).

1.3.3. The WF creates Mission Weather Products (MWPs) that fuse theater scale products with local mission requirements to enable the direct inject of weather impacts into warfighter planning and/or execution as required. The WF provides flight weather briefings and operational weather support for the 60 AMW, 349 AMW, and the Navy VQ-3 Detachment, if requested. The WF also provides flight weather briefings for transient aircrews IAW the WF duty priorities listed in **Table 1.1**.

1.3.4. TACC/XOW. TACC/XOW provides flight weather briefings and operational weather support for the 60 AMW's and the 349 AMW's IFM designated missions.

1.3.5. Travis AFB Installation Data Page (IDP). The 25 OWS and Travis AFB WF coordinate on and maintain the Travis AFB IDP. Data page details include TAF specification and amendment criteria, WWA thresholds, desired lead times, mission impacts, unit information, Joint Environmental Toolkit (JET) backup contacts and local outage backup information. Travis AFB IDP can be found at the following link: **https://25ows.us.af.mil/ows_unique/25data/moa/Travis_AFB_Data_Page.pdf**.

1.3.6. Eyes Forward & Collaboration. The Travis AFB WF acts as the eyes forward for the 25 OWS by relaying significant, time-sensitive meteorological information not found in coded meteorological reports to the assist in forecast operations.

1.3.7. Duty Priorities. All WF tasks cannot be accomplished simultaneously. Therefore, IAW **AFMAN 15-129** the WF has established the duty priorities listed in **Table 1.1** based on their order of relative importance to mission requirement. Since not all situations affecting operations at Travis AFB can be anticipated, WF personnel should use sound risk management principles to determine the need to recall additional personnel to assist in meeting surges in operations.

Table 1.1. 60 OSS/OSW Duty Priority Listing, Part 1.

Priority	Duties
1	Perform WF Emergency War Order (EWO) Tasks
2	Execute WF Evacuation
3	Respond to Aircraft/Ground Emergencies

Table 1.2. 60 OSS/OSW Duty Priority Listing, Part 2.

Priority	Duties
4	Respond To Pilot-to-Metro Service (PMSV) Contacts
5	Prepare and Disseminate Weather Watches, Warnings, and Advisories (WWAs)
6	Disseminate weather observation
7	Disseminate Urgent Pilot Reports (UUA/PIREPs)
8	Disseminate all PIREPs
9	Severe Weather Action Plan (SWAP) Operations
10	Disseminate Terminal Aerodrome Forecasts
11	Provide Flight Weather Briefings
12	Collaborate WPs with Supported Units
13	Perform MISSIONWATCH/METWATCH/Amend Weather Products
14	Provide Staff Briefings / Non-Standard WPs
15	Weather Functional Training
16	Accomplish Administrative Tasks

1.4. Hours of Operation.

1.4.1. WF. The WF remains open 24 hours a day, 7 days a week, in concurrence with airfield operating hours. In the event of airfield closure, the WF may close to match airfield hours. Staff services are available Monday-Friday 0800L-1600L or as required.

1.4.1.1. In the event of an airfield/WF closure, the WF technician is responsible for contacting the 25 OWS, Air Traffic Control (ATC), Airfield Management, and respective command posts (CPs). The weather technician does not leave the work center until all agencies have been notified.

1.4.1.2. During any WF closure, flight leadership will designate personnel to remain on "standby" status. Contact information for the standby forecaster is included below.

1.4.2. 25 OWS and TACC/XOW. Hours of operation are 24/7, 365 days a year.

1.4.3. A web-based aircrew-briefing terminal is located in the base operations flight planning room. This briefing terminal allows aircrews to self-brief or schedule a flight weather briefing from the 25 OWS.

1.5. Contact Information.

1.5.1. WF (707) 424-3003/5549 / DSN 837-3003/5549

1.5.2. WF Alternate Operating Location (707) 424-3003/2975/2110 / DSN 837-3003/2975/2110

1.5.3. Standby Forecaster Cell Phone (707) 639-7579

1.5.4. PMSV 271.1 MHz

1.5.5. TACC/XOW (618) 229-0353 / DSN 779-0353

1.5.6. 25 OWS (520) 228-6673 / DSN 228-6673

1.6. Business Continuity Rules. S upport to the installation and flying operations is susceptible to equipment and communication outages at the 25 OWS, TACC/XOW, and the WF. The WF participates in various wing, 25 OWS, and AMC exercises to maintain procedures for and proficiency at tasks necessary to ensure continuity of support.

1.6.1. WF Business Continuity Rules. In the event of a building evacuation, the WF relocates to their Alternate Operating Location (AOL) with contact numbers listed in **para. 1.5.2**.

1.6.1.1. WF members utilize evacuation checklists and standard operating procedures (SOPs) to expedite resumption of installation weather support. If access to observing sensing equipment readouts is lost, manual equipment is used to take observations.

1.6.1.2. Expect some services to be degraded (weather products, pilot briefings, etc.) due to facility limitations and loss of dedicated data services, including sensors and various data types (meteorological satellite (METSAT), radar imagery, etc.).

1.6.1.3. For flight safety reasons, the WF does not evacuate during exercises.

1.6.2. TACC/XOW Business Continuity Rules. IAW **AMCI 15-101**, if TACC/XOW loses the capability to provide flight weather briefing services, the WF assumes responsibility for briefing 60 AMW's and 349 AMW's IFM designated missions. If the WF is unable to provide the required support, briefing responsibility is transferred to the 25 OWS.

1.6.3. Weather Exercises. AMC weather units typically conduct coordinated exercises on the second Wednesday/Thursday of each month. AMC/A3AW issues Special Instructions (SPINS) outlining the functions and capabilities to be exercised. On a quarterly basis, the WF determines what capability needs to be exercised based on local training needs. As a courtesy, the WF coordinates timelines and potential impacts with supported units prior to all scheduled exercises.

Chapter 2

AIRFIELD SUPPORT

2.1. General. Airfield support includes those actions affecting the Travis AFB aerodrome (defined within a 5NM radius of the airfield) or Travis AFB as a whole. These functions include, but may not be limited to, weather observing, meteorological watch, and resource protection.

2.2. Observations. The FMQ-19 AMOS works in concert with JET to evaluate, prepare, and transmit weather observations for Travis AFB. IAW **AFMAN 15-111**, automated systems at Air Force and Army controlled airfields are to operate in full-automated mode to provide the official airfield observation. The three basic types of observations provided are Aviation Routine Weather Report (METAR), Aviation Selected Special Weather Report (SPECI), and Aviation Selected Local Weather Report (LOCAL). **Attachment 2** lists all SPECI criteria for Travis AFB. **Attachment 4** provides an example of a typical SPECI observation.

2.2.1. Point Of Observation. The official point where a weather observation is taken. On Travis AFB, the point of observation is where the FMQ-19 sensors are located (see **Attachment 9** for sensor group locations). During periods of augmentation from the primary site, the point of observation is marked by a North facing arrow directly outside of the P-4 flight line exit adjacent to the far end of the red carpet. When augmentation is required from the AOL, the official observation point is marked by a compass rose approximately 45 ft from the AOL flight line exit door.

2.2.2. Observing Location Limitations.

2.2.2.1. The FMQ-19 is properly sited and no limitations are currently noted.

2.2.2.2. When augmenting observations at the primary site (Bldg P-4), obstructions from the West to North may impact derived prevailing visibility. Additionally, observers may not hear thunder due to the level of noise on the flight line. Air Traffic Control (ATC) personnel supplement surface observations when required as identified in **para. 2.6** of this instruction.

2.2.2.3. When augmenting observations at the AOL, obstructions to the Northeast and West through Northwest may impact derived prevailing visibility. Additionally, observers may not hear thunder due to the level of noise on the flight line.

2.2.3. FMQ-19 Automated Observation. An automated observation is any observation evaluated, prepared, and transmitted by an observing system without human intervention. When operating in automated mode, the FMQ-19 determines sky condition based on an evaluation of sensor data gathered during the 30-minute period ending at the actual time of the observation. All other evaluated elements are based on sensor data that is within 10 minutes or less of the actual time of the observation.

2.2.4. FMQ-19 Augmentation. Augmentation is a method of having a position-qualified weather technician manually add or edit data to an observation generated by the FMQ-19. The two augmentation processes used are supplementing and backup.

2.2.4.1. Supplementing. Supplementing is the process of manually adding meteorological information to an observation generated by the FMQ-19 that is beyond the system's capability to detect and/or report. For example, the sensor cannot detect tornadoes or hail. WF personnel

supplement observations when the weather conditions in **AFMAN15-111 Table 5.1.** are observed or are forecast to occur within 1 hour.

2.2.4.2. Backup. Backup is a method of manually providing meteorological data and/or dissemination of an FMQ-19 generated observation when the primary automated method is not operational or unavailable due to sensor and/or communication failure. In the event of FMQ-19 malfunction or failure, WF personnel use manual observing procedures when performing backup operations. When required, the WF encodes and disseminates METAR and SPECI observations IAW **AFMAN 15-111**. All element entries are observed within 15 minutes of the actual time of the observation with the exception of wind gusts and squalls, which are reported only if they are observed within 10 minutes of the time of the observation. When backing up FMQ-19 wind sensors, the wind values will be reported as estimated. Unless a properly sited TMQ-53 is used to back-up the FMQ-19 pressure sensor, the pressure values will be reported as estimated.

Table 2.1. Mandatory Supplementary Weather Conditions.

Mandatory Supplementary Weather Conditions – Body of Report
Tornado (+FC) (Note 1)
Funnel Cloud (FC) (Note 1)
Waterspout (+FC) (Note 1)
Hail (GR) (local warning criteria only)
Volcanic Ash (VA)
Ice Pellets (PL)
Mandatory Supplementary Weather Conditions – Remarks Section of Report
Funnel Cloud (Tornadic Activity_B/E(hh)mm_LOC/DIR_(MOV)) (Note 2)
Snow Depth (Note 2)(only during airfield operating hours)
Tower Visibility (Note 3)
Notes: 1. The immediate reporting of funnel clouds takes precedence over any other phenomenon. 2. Remarks and Additive Data references are provided in AFMAN 15-111, **Attachment 3**. 3. Whenever tower visibility is reported by ATC personnel, and either surface visibility or tower visibility is less than 4SM and differs by at least one reportable value.

2.3. Terminal Aerodrome Forecasts (TAFs). Travis AFB TAFs are produced by WF and are disseminated IAW **AFI 15-128**, **AFMAN 15-124**, and **AFMAN 15-129.** TAFs are valid for 30 hours, apply to the area within 5NM of the Travis AFB airfield complex, and are issued at 0300, 1100, and 1900 Zulu time. A complete list of TAF specification and amendment criteria can be found on the Travis IDP at **https://25ows.us.af.mil/ows_unique/25data/moa/Travis_AFB_Data_Page.pdf**. **Attachment 3** lists forecast specification and amendment criteria. **Attachment 4** contains an example of a typical Travis AFB TAF.

2.4. RP Products. Special Weather Statements (SWS) and WWAs are special notices used to alert decision makers of hazardous weather occurring or the potential for hazardous weather to occur within the area of operations. **Attachment 4** provides an example of a Watch, Warning, and Advisory. Customer responses to WWAs are listed in **Attachment 5**.

2.4.1. SWS. A notice issued by the 25 OWS to assist with RP decisions. SWS advise of the potential for widespread hazardous weather conditions in a specified geographical region. The

WF utilizes the SWS to assist in maintaining situational awareness of environmental conditions that have the potential to negatively impact Travis AFB or the local flying area.

2.4.2. Weather Watch. A special notice to notify installation personnel and supported units of a potential for environmental conditions of such intensity as to pose a hazard to life or property. Watches are issued for the criteria defined in the Travis AFB IDP at https://25ows.us.af.mil/ows_unique/25data/moa/Travis_AFB_Data_Page.pdf and are valid for a 5NM radius from the center of the Travis AFB runway complex. Watches are issued for the criterion defined in **Table 2.2**.

2.4.2.1. Watches are standalone products based upon potential and are unaffected by warnings or advisories for the same phenomena.

2.4.2.2. Multiple watches may be in effect at the same time.

Table 2.2. Weather Watches, Part 1.

Watch Type	Criteria	Desired Lead Time
Tornado	Within 5NM	As potential warrants
Severe Thunderstorms	Winds ≥ 50 Knots or Hail ≥ 3/4 Inch	As potential warrants
Damaging Winds	≥ 50 Knots	As potential warrants
Freezing Precipitation	Any Intensity	As potential warrants

Table 2.3. Weather Watches, Part 2.

Watch Type	Criteria	Desired Lead Time
Large Hail	Hail ≥ 1/2 but < 3/4 inch	As potential warrants
Heavy Rain	≥ 2 Inch Accumulation in ≤ 12 Hours	As potential warrants
Lightning	Within 5NM	30 Minutes

2.4.3. Weather Warning. A special notice to inform installation personnel when an established weather condition of such intensity as to pose a hazard to life or property is occurring or is expected to occur. Warnings are issued for criteria defined in the Travis AFB IDP at https://25ows.us.af.mil/ows_unique/25data/moa/Travis_AFB_Data_Page.pdf and are valid for a 5NM radius from the center of the Travis AFB runway complex. Warnings are issued for the criterion defined in **Table 2.4**.

2.4.3.1. When multiple warning criteria are forecast to occur, technicians may issue either a single warning containing multiple warning criteria or a separate warning for each criterion. If a warning is issued for one weather criterion and it becomes necessary to warn for another weather criterion, a new warning, with a new number, is issued.

Table 2.4. Weather Warnings.

Warning Type	Criteria	Desired Lead Time
Tornado	Expected or Occurring Within 5NM	15 Minutes
Severe Thunderstorms	Winds ≥ 50 Knots or Hail ≥ 3/4 Inch	1 Hour
Moderate Thunderstorms	Winds ≥ 35 but < 50 knots or Hail ≥ 1/2 inch but < 3/4 inch	1 Hour
Damaging Winds	≥ 50 Knots	2 Hours
Strong Winds	≥ 35 Knots but < 50 Knots	1 ½ Hours
Freezing Precipitation	Any Intensity	1 Hour
Heavy Rain	≥ 2 Inch Accumulation in ≤ 12 Hours	1 Hour
Lightning	Within 5NM	Observed

2.4.3.2. Observed Weather Warnings. Lightning warnings are the only observed warning issued for Travis AFB. Lightning warnings are issued when lightning is observed within 5NM, either visually or via the National Lightning Detection Network. Lightning warnings are cancelled when lightning has not been observed within the past 15 minutes and radar indicates thunderstorms are no longer occurring within 5NM.

2.4.4. Weather Advisories. A notice to inform end users when an established environmental condition affecting operations is occurring or is expected to occur at Travis AFB. Forecast weather advisory criteria are defined in **Table 2.5** and observed weather advisory criteria are defined in **Table 2.6** Observed weather advisory criteria are also defined in the Travis AFB IDP at **https://25ows.us.af.mil/ows_unique/25data/moa/Travis_AFB_Data_Page.pdf**.

Table 2.5. Forecast Weather Advisory.

Criteria	Forecast/Observed
Surface Winds ≥ 25 Knots but < 35 Knots	30 Minutes

Table 2.6. Observed Weather Advisories.

Criteria	Forecast/Observed
Low-Level Wind Shear	Observed
Crosswinds ≥ 31 kts with wet or dry runway (Note 1)	Observed
Crosswinds ≥ 27 kts with wet or dry runway (Note 1)	Observed
Crosswinds ≥ 19 kts with wet runway (Note 1)	Observed
Crosswinds ≥ 12 kts with wet runway (Note 1)	Observed
Moderate or greater Icing below 10,000ft AGL	Observed
Moderate or greater Turbulence below 10,000ft AGL	Observed
Lightning within 25NM of Travis AFB	Observed
Lightning within 10NM of Travis AFB	Observed
Notes: 1. Crosswinds are calculated based on the maximum observed wind speed (to include gusts) and worst-case observed direction, including variability.	

2.4.5. WWA Numbering Scheme. WWAs are numbered consecutively by identifying the type of weather message (watch, warning, or advisory) followed by a five-digit number. The first two numbers indicate the current month while the second three numbers indicate the sequence number. For example, the message "Weather Warning 02-005" means the month is February (02) and this is the fifth (005) warning issued in the month. The message "Weather Advisory 12-013" means the month is December (12) and this is the thirteenth (013) advisory issued in the month. An example of an observed weather advisory is contained in **Attachment 4**.

2.4.6. WWA Upgrades/Downgrades. An upgrade is a change to an active WWA resulting from adding additional WWA phenomenon or an increase in phenomenon intensity that crosses to a higher threshold (e.g., winds increase from 35 knots to 50 knots). A downgrade is a change to an active WWA resulting from removing WWA phenomenon or a decrease in phenomenon intensity that crosses to a lower threshold (e.g., hail decreases from ¾ inch to ½ inch).

2.4.7. WWA Amendments. Amendments are issued when an active WWA no longer adequately describes a phenomenon's expected occurrence. All amendments are issued with a new WWA number and should clearly state what affect the new WWA has on any previously issued notices.

2.4.8. WWA Extensions. Extensions are issued when a phenomenon's occurrence is expected to last longer than originally forecast. Extensions are issued prior to the expiration of the original WWA utilizing the same WWA number. Extensions should clearly state what effect they have on any previously issued notices.

2.4.9. WWA Cancellation. WWA are canceled when the weather phenomena is no longer occurring or expected to occur. WWA not extended or canceled automatically expire at the end of the valid period. Observed advisories are canceled when the criteria has not occurred in the last 30 minutes with the exception of observed lightning advisories and warnings. Lightning advisories and warnings are cancelled when lightning has not been observed within the past 15 minutes and radar indicates that thunderstorms are no longer occurring within the valid range.

2.5. Dissemination Process.

2.5.1. Observations. Observations taken by either the FMQ-19 or the weather technician are disseminated via JET. When JET is nonoperational, the WF relays observations to the following local organizations in order of priority listed in **Table 2.6**.

Table 2.7. Notification Priority.

1. Tower commercial (707) 424-4694
2. RAPCON (707) 424-5111
3. 60 AMW/CP commercial (707) 424-5517
4. Airfield Management commercial (707) 424-2836
5. 25 OWS commercial (520) 228-7655

2.5.2. TAFs. The Travis AFB WF disseminates TAFs via JET. If JET is nonoperational, the WF disseminates the TAF to ATC and 60 AMW/CP via telephone, fax, or e-mail.

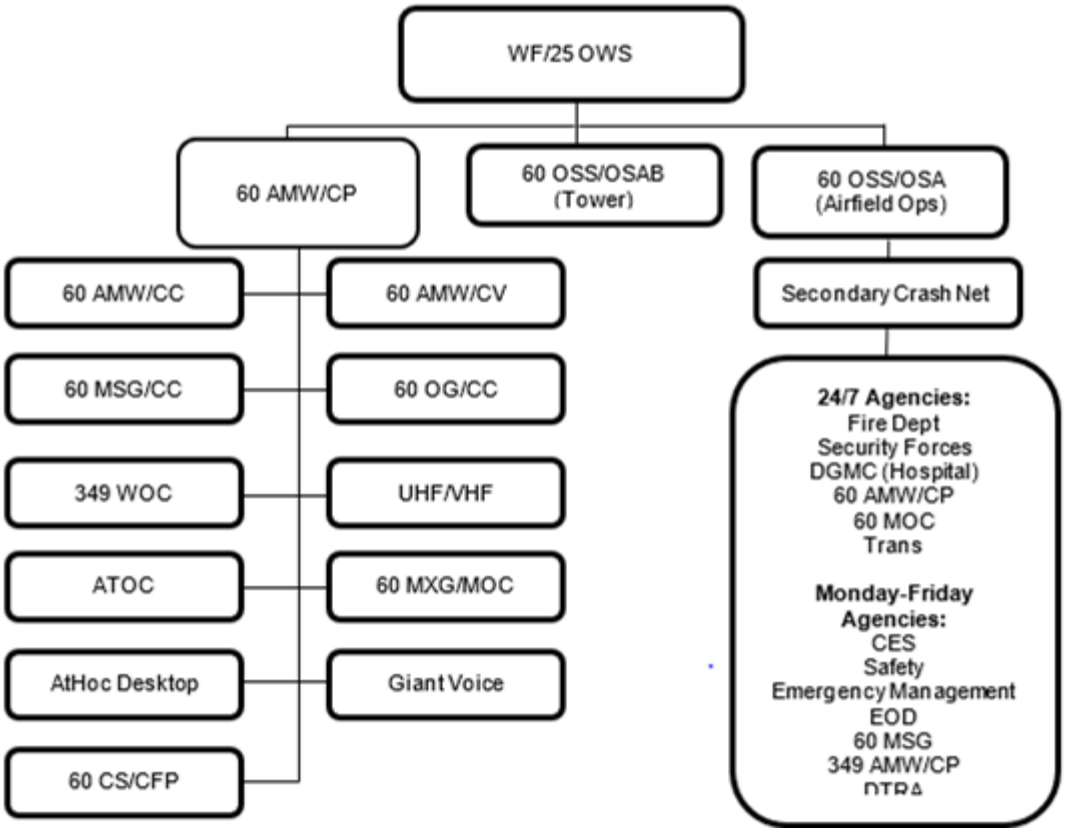
2.5.3. Special Weather Statements (SWS). Special Weather Statements give advanced notice to leadership of expected hazard and impacts. 25 OWS or WF transmits SWSs to AMW leadership via e-mail.

2.5.4. WWAs. The WF or the 25 OWS enters WWAs into JET for dissemination to ATC, 60 AMW/CP, and 60 OSS/OSAA Airfield Management. If JET is out-of-service, the WF or the 25 OWS disseminates WWAs via back-up calls. Upon notification, units identified in **Figure 2.1** should further disseminate *all* WWAs using the pyramid notification scheme. In addition, the 60 AMW/CP disseminates all WWAs via e-mail and/or AtHoc.

2.5.4.1. Lightning Warnings. All lightning warnings are disseminated by the 60 AMW/CP to the base populace via the Giant Voice system.

2.5.4.2. Tornado Warnings. The 60 AMW/CP has the primary responsibility for sounding the base siren for a tornado warning issuance.

Figure 2.1. Weather Alerting Pyramid.



2.6. Cooperative Weather Watch (CWW). The WF and ATC have established a CWW IAW **AFMAN 15-111**. The agreement outlines each unit's responsibilities when specific meteorological phenomena are observed. Of primary concern is the report of tower visibility differing from the prevailing surface visibility, local PIREPs**,** and any occurrence of previously unreported weather conditions that could affect flight safety or be critical to the safety or efficiency of other local operations and resources. WF technicians and ATC personnel should thoroughly understand and be able to execute the CWW agreement. Specifically:

2.6.1. Travis AFB Tower personnel:

2.6.1.1. Notify the weather technician when the observed tower prevailing visibility decreases to less than or increases to equal or exceed 1, 2, 3, or 4 miles, and is different than the surface prevailing visibility.

2.6.1.2. Notify the weather technician whenever there are operationally significant changes to tower prevailing visibility and the tower visibility is different than the reported surface prevailing visibility.

2.6.1.3. Notify the weather technician when PIREPs are received of previously unreported weather conditions that could affect flight safety or be critical to the safety and efficiency of other local operations and resources. PIREP information should be relayed to weather personnel no later than 5 minutes after receipt.

2.6.1.4. Notify the weather technician when tower personnel detect significant weather phenomena (i.e., lightning, precipitation, low ceilings, etc.) and the phenomena is not reflected in the current surface weather observation.

2.6.2. Travis AFB weather technicians:

2.6.2.1. Re-evaluate the surface weather observation when tower personnel report significant weather phenomena not reflected in the current observation.

2.6.2.2. Re-evaluate the surface prevailing visibility, as soon as practicable, upon notification that the control tower prevailing visibility is different than the surface prevailing visibility and either the surface visibility or tower visibility is less than 4 miles. **Note:** Tower visibility is included in remarks when received (reference **Figure 2.1**).

2.6.2.2.1. Use control tower values of prevailing visibility as a guide in determining the surface visibility when the view of portions of the horizon is obstructed by buildings, aircraft, etc.

2.6.2.2.2. Include a tower visibility remark in the next METAR or SPECI when either the surface prevailing visibility or the control tower prevailing visibility is less than 4 statute miles (6000 meters) and they differ by one or more reportable values.

2.6.3. ATC Limited Observation Training. The WF oversees the Travis AFB Limited Observation Training Program. ATC personnel requiring training should contact the WF Chief at DSN 837-7090/Commercial (707) 424-7090 to schedule an appointment. To satisfy Limited Observation Training requirements, personnel are required to pass a written exam and receive an orientation of the weather facilities.

2.7. PMSV Support. Weather information is available via PMSV on frequency 271.1 MHz. The duty forecaster monitors PMSV traffic for all aircraft contacts. For aircraft outside the range of our PMSV system, the Travis AFB WF or 25 OWS can provide PMSV support through a phone patch to the 60 AMW/CP (DSN 837-5517/Commercial (707) 424-5517). PMSV outages are discussed in **para. 5.3.2**.

2.8. Emergency Action(s) Response.

2.8.1. Aircraft Mishap. When notified of an aircraft mishap, the WF initiates a save of applicable data used in the development of any weather products provided. Archived weather data is available to investigating agencies upon request.

2.8.1.1. If the WF provided the MWP, the WF notifies the 25 OWS Senior Duty Officer of all aircraft mishaps as soon as possible after notification of the event. The WF coordinates with the 25 OWS to save all applicable data and products. If products from other OWSs were used, the WF coordinates with all applicable OWSs to ensure the correct data is saved. OWSs save sufficient data to fully reconstruct environmental conditions before and after the mishap.

2.8.1.2. If an OWS, TACC/XOW, or another WF provided the MWP, they are responsible for conducting the data save in coordination with any other Air Force Weather units involved.

2.8.2. SWAP. The SWAP ensures sufficient manpower is available to meet the increased demand for timely weather information during significant weather events. The WF duty forecaster initiates the SWAP when any of the criteria in **Table 2.7** are forecast to occur.

Table 2.8. Conditions Requiring SWAP Activation.

SWAP ACTIVATION Criteria
Watch/Warning for Tornado(s)
Watch/Warning for Severe Thunderstorms w/Hail ≥ 3/4 Inch
Watch/Warning for Damaging Winds ≥ 50 kts
Watch/Warning for Heavy Rain ≥ 2 Inches in 12 hrs
Watch/Warning for Freezing Precipitation

2.8.3. Chemical, Biological, Radiological, Nuclear, and High-yield Explosive (CBRNE) Response. The WF works closely with Emergency Management (EM) to ensure supported commanders receive the relevant information needed to obtain a timely and accurate picture of the environmental situation. Upon request, the WF provides EM:

2.8.3.1. Surface observations and/or alphanumeric forecasts representative of the location and time of the CBRNE event.

2.8.3.2. The appropriate Chemical Downwind Messages obtained from the 25 OWS, amended as necessary to ensure that it is representative of forecast conditions.

Chapter 3

MISSION SERVICES

3.1. General. The WF, 25 OWS, and TACC/XOW support the Travis AFB flying and non-flying missions. This chapter focuses on routine products and services provided to aviation units.

3.2. Flying Missions. The WF and the TACC/XOW provide weather support to the flying units listed in **Attachment 6**. The bulk of flying operations at Travis AFB are supported by the WF and TACC/XOW supports IFM designated missions.

3.3. MWP. MWPs fuse theater scale products with local mission requirements enabling the direct inject of weather impacts into warfighter planning and/or execution. The result is a product designed to provide timely, accurate, and relevant environmental information for planning and execution. MWPs should be horizontally consistent with (but not necessarily mirror) products issued by any OWS and the 557th Weather Wing (557 WW).

3.3.1. GDSS Weather Briefings. GDSS is the primary command and control (C2) system used by AMC owned/gained flying units to obtain weather briefings. GDSS provides a unique product that incorporates the requirements of all Travis AFB-based flying units into a common format. Updates for local missions can be obtained by contacting the WF at DSN 837-3003/Commercial (707) 424-3003. Updates for IFM missions can be obtained by calling TACC/XOW at DSN 576-0353/ Commercial (618) 229-0353.

3.3.2. WF MWP. The WF MWP is used to provide standardized weather briefings in the event of a GDSS outage, the WF may also use the WF MWP when executing TACC Business Continuity Rules. An example WF MWP is in **Attachment 8**.

3.3.3. Verbal MWP. Aircrews flying local missions may request a verbal briefing if no significant weather concerns are forecasted for their mission time. The information is recorded on the verbal briefing log maintained in the WF.

3.3.3.1. The Travis AFB Mission Execution Forecast (MEF). The MEF serves as the Takeoff and Landing Data (TOLD) product for pilots. TOLD provides a forecast of expected conditions over at Travis AFB for an 18-hour period, divided into hourly increments. It includes local area TAFs, solar/lunar data, airfield hazards, climb winds, and mission limiting thresholds for 8 other divert airfields. **Attachment 7** provides an example of the MEF.

3.3.3.2. The MEF is updated NLT 11 Zulu and 1900 Zulu daily or with any major updates to the TAF. The Travis AFB Weather MEF product is included into all mission briefs in GDSS and is also posted at: **https://eim2.amc.af.mil/org/60oss/weather/default.aspx**

3.4. MISSIONWATCH. MISSIONWATCH is a deliberate process for monitoring terrestrial weather or the space environment for specific mission-limiting, environmental factors that may adversely impact missions in execution. The MISSIONWATCH process is intended to identify previously unidentified environmental threats and alert decision-makers at the operational unit and/or airborne mission commanders, enabling dynamic changes to mission profiles that may mitigate the environmental threat and optimize the chance of mission success.

3.4.1. WF Briefed Sorties. The WF utilizes a continuous MISSIONWATCH process to validate that MWPs accurately reflect environmental conditions. When MWPs differ from observed conditions to the extent that it has the potential to impact operations, the WF coordinates MWP

amendments/updates with the appropriate OWS. Additionally, when previously unforecast weather conditions expose a mission to potential risk, the WF directly updates the 60 AMW/CP. The 60 AMW/CP retains responsibility for relaying updated weather information to the aircrew.

3.4.2. TACC/XOW Briefed Sorties. TACC/XOW performs MISSIONWATCH and weather risk assessment for all TACC/XOW briefed sorties. TACC/XOW uses the Weather Threat Assessment (WTA) to relay information on missions considered "at risk" based on the Operational Risk Management Thresholds identified in **AMCI 15-101**. Users can access the WTA at **https://tacc.us.af.mil/?action=WTAMain**.

3.4.2.1. The 60 AMW/CP automatically receives WTA notifications for wing-specific missions through a subscription service. Upon receipt of weather threat notification, the AMW/CP notifies appropriate Squadron Operations Centers/aircrews to pass along weather threats and instruct the aircrew to contact TACC/XOW to mitigate the threat.

3.4.2.2. The WF monitors GDSS for situational awareness on TACC/XOW briefed missions involving 60 AMW and 349 AMW assets. The WF coordinates with TACC/XOW whenever observed weather conditions deviate significantly from the published WTA.

3.5. Post-Mission Analysis/Feedback. Aircrews should contact the WF or TACC/XOW with post-mission information and/or follow-up support. The WF and TACC/XOW utilize customer feedback to improve internal processes and enhance training, forecast proficiency, and product accuracy. Formal/informal feedback methods include:

3.5.1. Completion of WF feedback worksheet or TACC/XOW feedback solicitation email.

3.5.2. Phone call or an e-mail to the WF or TACC/XOW.

3.5.3. Face-to-face feedback after briefing and/or mission completion.

3.6. Transient Aircrew Support. Transient aircrews may submit requests for flight weather briefings (175-1s) in person or via phone using the contact information listed in **para. 1.5** If the WF workload is such that they are unable to support transient crew requests (see **Table 1.1** for WF duty priorities), the WF may direct aircrews to contact the 25 OWS briefing cell for flight weather information. The 25 OWS briefing cell can be reached at DSN 228-6598/6599, commercial (520) 228-6598/6599, or via web access from the aircrew briefing terminal located in the flight planning room.

3.7. Aero Club Activities. The WF provides flight weather briefings to Aero Club members performing official Air Force operational duties (i.e. Civil Air Patrol and Initial Flying Training Programs). The WF provides or arranges briefings when such Aero Club flights are in a transient status through the appropriate OWS or Flight Service Station. The WF advises Aero Club members performing official flight duties of the OWS web page request process and self-briefing capabilities.

3.8. Non-Flying Missions. The WF and 25 OWS support various non-flying missions (e.g., Wg Picnic, change of command ceremonies, Morale Welfare and Recreation) through RP (WWAs). Specific support to non-flying missions is identified in **Chapter 4**.

3.9. Space Weather Impacts. Travis AFB has a wide-variety of operations affected by various space-weather parameters (High Frequency and Ultra High Frequency communication, radar, Global Positioning System communications, etc.). The WF and TACC/XOW provide space impacts on their MWPs. More detailed space environmental situational awareness products are

available at **https://weather.af.mil/confluence/display/AFWWEBSTBT/CONUS+Regional+Space+Weather**

Chapter 4

STAFF SERVICES

4.1. General. Staff services are typically accomplished by WF leadership and include meteorological functions (briefings), ensuring the WF is trained and equipped for day-to-day operations, and cultivating relationships with base agencies to ensure WF support is optimal.

4.2. Staff Meteorological Functions. Staff meteorological functions aid leadership in identifying and understanding specific weather and environmental impacts. Specialized weather information can be provided to support any non-flying mission upon request. The WF is available to assist commanders in determining weather support requirements and impacts to operations. Non-governmental agencies should request weather information and support through 60 AMW/PA. Examples of staff meteorological functions provided are:

4.2.1. Staff Briefings. Staff weather briefings for 60 AMW, 349 AMW (Wing stand up) and 60 OG (group stand-up) are provided upon request. Standard information includes a 5-day Travis AFB weather outlook with a focus on any affected Wing events or missions. Additional information is provided as it pertains to operations and safety, or at the request of OSS, OG, or AMW leadership.

4.2.2. Installation Control Center (ICC)/Crisis Action Team (CAT) Briefings. The WF provides weather support as required for ICC/CAT briefings. This includes real-world emergency, exercise, and deployment briefings. Each briefing is tailored to provide the appropriate weather intelligence required by the 60 AMW and 349 AMW leadership.

4.2.3. Instrument Refresher Course (IRC) Briefings. IAW **AFMAN 11-210**, *Instrument Refresher Program (IRP),* computer-based training is available for the weather portion of the briefing. If requested, the WF can provide a briefer to discuss more detailed local weather effects and impacts. This briefing includes airfield and mission services, WF capabilities, resource protection, seasonal/regional weather and space weather impacts (when applicable).

4.2.4. Pre-deployment Planning Briefings. The WF provides pre-deployment weather briefings as requested. Briefing content is tailored to meet customer requirements. For example, an aviation unit receives weather impacts at the deployed location on their flying mission, in addition to the standard surface weather information usually presented to ground units. A ground-based unit may receive a briefing on surface temperatures, wind speed, potential for blowing sand and dust, precipitation and pertinent environmental factors.

4.2.5. Climatology Services. The WF can provide a wide variety of climatology products upon request. Example products include, but are not limited to, historical surface observations, long-range outlooks, global cloud cover, and upper-level wind climatology.

4.3. Staff Integration Functions. In addition to leadership and management of unit activities, these unit members also function as a direct interface with the supported unit commander and staff, and provide direct support to command, control and planning functions. Specific procedures/functions for integration with base agencies are outlined below.

4.3.1. 60 AMW/XP and 349 AMW/XP (Plans and Programs). The WF assists in periodic exercises tailored to upcoming seasonal weather or other environmental concerns and educates base agencies on the purpose and applicability of WWAs.

4.3.2. 60 AMW/CP and 349 AMW/CP. The WF notifies the CPs whenever the WF primary facility is evacuated and/or the AOL is activated.

4.3.3. 60 AMW/PA. The WF provides tours of the WF facility for community groups and others when coordinated by Public Affairs.

4.3.4. 60 OSS/OSAA (Airfield Management). The OWS or WF provides notification of all WWAs via JET, telephone, e-mail, or in-person during airfield hours of operations.

4.3.4.1. The WF notifies OSAA whenever the WF primary facility is evacuated and/or the AOL is activated.

4.3.4.2. WF leadership participates as a member of the Airfield Operations Board (AOB) as directed in **AFMAN 13-204 Vol I**, *Airfield Operations Procedures and Programs*.

4.3.5. 60 OSS/OSAB (Air Traffic Control).

4.3.5.1. The WF provides notification of all WWAs via JET, telephone, or e-mail during airfield hours of operations.

4.3.5.2. The WF notifies OSAB whenever the WF primary facility is evacuated and/or the AOL is activated.

4.3.5.3. The WF notifies OSAB of all outages prior to contacting any maintenance agency.

4.3.6. 60 CES. The WF provides a monthly climatology report, accessible via the WF SharePoint page.

4.3.7. All Supported Flying Units (60 AMW, 349 AMW, and Navy VQ-3 Detachment). The WF provides services as outlined throughout this publication.

4.4. Reciprocal Support.

4.4.1. 60 AMW and 349 AMW CPs.

4.4.1.1. Ensure dissemination of all WWA as outlined in **Chapter 2** of this document.

4.4.1.2. Immediately notify the WF forecaster-on-duty of any aircraft or ground mishaps (weather-related or not) requiring OPREP-3 reporting or local reporting requirements IAW **AFMAN 10-206**.

4.4.1.3. Subscribe to and monitor the WTA notifications for 60 AMW and 349 AMW missions.

4.4.1.4. Upon notification, by either the WF or 25 OWS, of the potential for severe weather, utilize applicable Quick Reaction Checklists (QRCs) to alert wing leadership and various base agencies.

4.4.2. 60 AMW/CP. Activate base emergency sirens when a tornado WARNING is issued. Disseminate all warnings via email and/or AtHoc.

4.4.3. 60 AMW/PA. Coordinate requests for weather information from non-DoD agencies and tours of WF facilities with the WF Chief.

4.4.4. 60 OSS/OSAA.

4.4.4.1. Notify WF personnel of in-flight, ground emergencies, or mishaps and termination via the secondary crash network.

4.4.4.2. Disseminate weather warnings and advisories as outlined in **Chapter 2** of this instruction.

4.4.4.3. Notify WF of runway surface condition changes, as this impacts crosswind thresholds.

4.4.4.4. Take appropriate Notice to Airman (NOTAM) action when notified by the WF of changes in the operational status of weather equipment or evacuation of the WF to the alternate operating location lasting over 24 hours.

4.4.4.5. Submit WF updates to flight publications when required.

4.4.5. 60 OSS/OSAB.

4.4.5.1. Participate in and follow the procedures outlined in CWW program.

4.4.5.2. Notify the WF of all changes in active runway.

4.4.5.3. Notify the WF of any light setting changes on the high-intensity runway lights.

4.4.5.4. Relay pilot weather reports to weather personnel within 5 minutes of receipt.

4.4.5.5. Provide control tower orientation training for weather personnel.

4.4.5.6. Initiate/respond to radio checks to ensure proper PMSV operation.

4.4.5.7. Monitor Travis PMSV frequency (UHF 271.1. MHz) during WF outages, in accordance with established duty priorities.

4.4.6. 60 OSS/OSAM (Radar and Weather Systems).

4.4.6.1. Provide, coordinate, or arrange for the installation, maintenance, and repair of all weather communication and meteorological sensing equipment except for the communication and meteorological equipment maintained by contract.

4.4.6.2. Coordinate with the weather technician prior to performing maintenance on weather communications or equipment. Ensure routine maintenance does not degrade METWATCH and/or MISSIONWATCH performed by the WF during periods of inclement weather.

4.4.6.3. Utilize the restoration priorities for weather communications and meteorological sensing equipment outlined in this document.

4.4.6.4. Notify the responsible service agents for weather communications and meteorological sensing equipment outages.

4.4.6.5. Perform necessary follow-up actions as required until full service is restored.

4.4.6.6. Ensure established maintenance response times are met.

4.4.6.7. Ensure a 24-hour point of contact for reporting outages and assigning job control numbers is available.

4.4.6.8. Perform AN/TMQ-53 Tactical Meteorological Observing System pressure sensor calibrations annually, or when comparisons between the tactical pressure sensor and aircraft altimeter (or similar reliable sensors) indicate that calibration is warranted.

4.4.7. 60 CS.

4.4.7.1. Maintain and administer JET IAW the memorandum of agreement between 24 AF, AF Director of Weather, and the JET Program Management Office.

4.4.7.2. Coordinate with off-base agencies to repair off base lines.

4.4.7.3. Ensure weather data and telephone circuits are assigned repair priorities.

4.4.7.4. Perform necessary follow-up actions as required until full service is restored.

4.4.7.5. Ensure a 24-hour point of contact for reporting outages and assigning job control numbers is available.

4.4.8. 60 CS/SCOT. Provide or arrange for maintenance and repair of the PMSV radio system.

4.4.9. 60 AMW/SE. Request a Travis AFB WF briefer for seasonal weather briefings and provide 2 weeks advance notice when possible.

4.4.10. 60 CES. Contact WF Leadership to request climatological data and specialized support for projects on Travis AFB.

4.4.11. 60 SFS. Promptly inform the WF of any hazardous weather reported by Security Forces personnel (e.g., tornado, hail, etc.).

4.4.12. All Supported Flying Units (60 AMW, 349 AMW, and Navy VQ-3 Detachment).

4.4.12.1. Notify the WF of current and planned weather alternates and any special considerations affecting duration of mission (i.e., weather categories, exercise/deployment considerations, etc.).

4.4.12.2. Notify the WF of required additional support as soon as it becomes known, to include monitoring of alternate observations/forecast and tracking of weather conditions affecting local flying operations.

4.4.12.3. Provide timely notification of changes to scheduled operations affecting weather support requirements as soon as the change is identified.

4.4.12.4. Provide PIREPS either directly to the WF or through the PMSV or tower, or Supervisor of Flying (SOF).

4.4.12.5. Provide feedback on weather briefings via e-mail or survey to the WF or TACC/XOW.

4.4.12.6. Provide the WF a minimum of 2 weeks advanced notification of any requirement for training conducted by the WF or any changes in requirements to previously scheduled weather training.

4.4.13. Airfield Management Flight Information Publication (FLIP) Manager. Submit FLIP updates provided by the WF to Air Force Flight Standards Agency (AFFSA)/XOF.

4.4.14. 621st Contingency Response Wing (CRW). The ranking CRW weather forecaster serves as the point of contact for all weather related CRW activities, and should:

4.4.14.1. Collaborate with WF for tactical weather equipment proficiency training at least 2 weeks in advance.

4.4.14.2. Provide back-up briefing support to WF Staff Services function during periods of limited manning (exercises, deployments).

4.4.15. 60 AMDS/SGPB (Bioenvironmental Flight). Provide the base populace with the Wet Bulb Globe Temperature (WBGT) as required.

4.4.16. All Weather Support Recipients. All recipients should notify the WF when new weather support requirements are identified or when changes to current weather support is deemed necessary.

Chapter 5

WEATHER EQUIPMENT

5.1. General. This chapter provides a brief description of the meteorological and communications equipment used by the WF. Additionally, it provides information on backup systems, maintenance, and restoring priorities.

5.2. Meteorological Sensing. The WF uses the FMQ-19 and WSR-88D weather radar to determine the current state of the atmosphere. These critical systems provide customers the most timely, accurate, and relevant weather intelligence possible. **Note:** TMQ-53 is a tactical automated observing system that is used by the WF during contingency and exercise operations. The TMQ-53 provides a capability that is very similar to the FMQ-19.

5.2.1. FMQ-19. The FMQ-19 samples, measures, and reports: temperature, wind speed and direction, visibility, cloud base height and amount of coverage, pressure, liquid equivalent precipitation accumulation, and ice accretion during freezing precipitation. These measurements are processed to create properly formatted, fully automated observations that comply with applicable reporting standards and protocols defined by the World Meteorological Organization (WMO), Federal Aviation Administration (FAA), National Weather Service (NWS), and military reporting standards.

5.2.2. Gibson Ridge Software (GRS). The WF utilizes the GRS applications to access WSR-88D, Weather Surveillance Radar (i.e., NEXRAD) data. This software allows technicians to observe a large variety of both meteorological and non-meteorological phenomena ranging from sunrise spikes to detailed information on complex thunderstorm circulations. Weather technicians routinely incorporate the latest radar information into all mission execution forecasts and RP products.

5.3. Communications Equipment. The following systems are the backbone of the WF communications network:

5.3.1. JET. As discussed in **para. 2.5** of this instruction, JET is the primary system for disseminating forecast, observations, and WWAs. Telephones are used as a backup for key aircraft controlling agencies.

5.3.2. PMSV Radio. The PMSV Radio (271.1 MHz) allows the WF to communicate with aircrews, both on the ground and in the air. When the Travis AFB PMSV is out-of-service, aircrews can contact TACC/XOW at (618) 229-0353 / DSN 779-0353 or the 25 OWS at (520) 228-6598/6599 / DSN 228-6598/6599 to acquire weather information. Aircrews may also obtain weather information via phone patch from the WF, TACC/XOW, or the 25 OWS.

5.3.3. Phones/Hotlines. Phones and hotlines serve primarily for rapidly passing along critical, time-sensitive information, as well as to serve backup services.

5.3.4. Local Area Network (LAN). The LAN is the WFs primary way of monitoring current weather conditions and performing MISSIONWATCH. The WF also utilizes the LAN to obtain the majority of the weather model data used to determine the forecast state of the atmosphere.

5.4. Maintenance.

5.4.1. Organizations providing preventive maintenance and repair of weather and communications equipment are listed in **Table 5.1**.

Table 5.1. Equipment Maintenance List.

Organization	Equipment
60 OSS/OSAM (Radar and Weather Systems)	FMQ-19
557 WW Fielded Systems Support Center	JET
60 CS/SCOV (Telephone Systems)	Phones/Hotlines
60 CS/SCOI (Network Maintenance)	LAN/Internet Connectivity

5.4.2. Restoral Priorities. Priorities for restoring critical systems exist in the event of natural disasters or any other anomaly, simultaneously impacting systems base-wide. Significant indicates a situation where the equipment is completely inoperative, while minimal means the equipment is in limited operation. The priorities for weather equipment are listed in **Table 5.2** below (priorities may be adjusted based on forecasted weather). **Note:** If any sensor necessary to report ceilings, visibility, winds, altimeter setting, and/or discontinuity group sensor is inoperable, report outage as significant.

Table 5.2. Equipment Restoral Priorities.

Equipment	Organization	Response Times Significant/Minimal
PMSV Radio	60 CS/SCOT	Immediate/24 hours
FMQ-19	60 OSS/OSAM	Immediate/24 hours
LAN/Internet Connectivity/Phones/Hotlines/JET	60 CS/SCOI	Immediate/12 hours

5.5. Building Power. In the event of a commercial power interruption, Bldg 4 automatically switches to generator backup power.

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Commander, 60th Air Mobility Wing

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

DAFI 10-2501, *Emergency Management (EM) Program*, 17 March 2022

AFMAN 11-2KC-10V3, *KC-10 Operations Procedures*, 11 May 2022

AFMAN 11-2C-5V3, *C-5 Operations Procedures*, 15 April 2019

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AFI 11-208 IP*, Department of Defense Notice to Airmen (NOTAM) System,* 12 February 2018

AFMAN 13-204V1, *Airfield Operations Procedures and Programs*, 21 July 2020

AFI 15-128, *Weather Force Structure*, 20 June 2019

AFMAN 11-202V3, *Flight Operations*, 09 January 2022

AFMAN 10-206, *Operational Reporting,* 31 August 2020

AFMAN 11-210, *Instrument Refresher Program (IRP)*, 20 December 2021

AFMAN 15-111, *Surface Weather Observations*, 12 March 2019

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AFI 15-127, *Weather Training*, 26 January 2021

AFMAN 15-129, *Air and Space Weather Operations,* 9 July 2020

Abbreviations And Acronyms

AFI—Air Force Instruction

AFMAN—Air Force Manual

AFB—Air Force Base

AFFSA—Air Force Flight Standards Agency

AGL—Above Ground Level

AOL—Alternate Operating Location

AMC—Air Mobility Command

AMOS—-Automated Observing System

AMW—Air Mobility Wing

ARS—Air Refueling Squadron

AS—Airlift Squadron

ATC—Air Traffic Control

CAT—Crisis Action Team

CB—Cumulonimbus

CBRNE—-Chemical, Biological, Radiological, Nuclear, and High-yield Explosive

CC—Commander

CES—Civil Engineering Squadron

CONUS—Continental United States

CP—Command Post

CRW—Contingency Response Wing

CS—Communications Squadron

CWW—Cooperative Weather Watch

EM—Emergency Management

EWO—Emergency War Orders

FLIP—Flight Information Publication

FTU—Formal Training Unit

GDSS—Global Decision Support System

HF—High Frequency

IAW—In Accordance With

ICAO—International Civil Aviation Organization

ICC—Installation Control Center

IRC—Instrument Refresher Course

JET—Joint Environmental Toolkit

KT—Knots

LAN—Local Area Network

METAR—Meteorological Terminal Aviation Routine Report

METSAT—Meteorological Satellite

METWATCH—Meteorological Watch

MOV—Moving

MOVD—Moved

MWP—Mission Weather Product

NOTAM—Notice to Airmen

NWS—National Weather Service

OG—Operations Group

OHD—Overhead

OL-D—Operating Location-D

OPR—Office of Primary Responsibility

OSAA—Airfield Management

OSAB—Air Traffic Control

OSAM—Radar and Weather Systems

OSS—Operations Support Squadron

OWS—Operational Weather Squadron

PA—Pressure Altitude

PIREP—Pilot Report

PK WND—Peak Wind

PMSV—Pilot-to-Metro Service

RDS—Records Disposition Schedule

RP—Resource Protection

RVR—Runway Visual Range

RWY—Runway

SE—Safety Office

SFS—Security Forces Squadron

SM—Statute Mile

SOF—Supervisor of Flying

SOP—Standard Operating Procedure

SPECI—Special Weather Observation

SWAP—Severe Weather Action Plan

SWS—Special Weather Statement

TACC—Tanker Airlift Control Center

TAF—Terminal Aerodrome Forecast

TCU—Towering Cumulus

TWR—Tower

UFN—Until Further Notice

UHF—Ultra High Frequency

UNKN—Unknown

VFR—Visual Flight Rules

VHF—Very High Frequency

VIS—Visibility

WF—Weather Flight

WSHFT—Wind Shift

WSR-88D—Weather Surveillance Radar, 1988 Doppler

WWA(s)—Warnings, Watches, and Advisories

Attachment 2

SPECIAL WEATHER OBSERVATION CRITERIA

A2.1. A Special weather observation is taken and disseminated for listed criteria:

A2.1.1. Visibility. When the prevailing visibility decreases below or, if below, increases to equal or exceeds any of the values listed below:

Table A2.1. Visibility Thresholds.

Visibility (Statute Miles)	**3**	**2**	**1 7/8**	**1 ¾**	**1 5/8**	1 ½	**1 3/8**	**1 ¼**	**1**	**7/8**	**¾**	**½**	¼
	Note: Items in bold/underline indicate criteria found in the high and low altitude FLIPs.												

A2.1.2. Ceiling. When the ceiling goes below or, if below, increases to equal or exceeds any of the values listed below:

Table A2.2. Ceiling Levels.

Height (feet)	3,000	2,000	1,500	**1,200**	**1,100**	1,000	**800**	**700**	**600**	**500**	300	**200**	100
	Note: Items in bold/underlined indicate criteria found in the high and low altitude FLIPs												

A2.1.3. Sky Condition. A layer of clouds (does not have to be a ceiling) or obscuring phenomena aloft is observed below 1,400 feet and no layer was reported below this height in the previous METAR or SPECI.

A2.1.4. Wind.

A2.1.4.1. Shifts. A directional change of 45 degrees or more in less than 15 minutes with sustained winds of 10 knots or more throughout is considered a wind shift.

A2.1.4.2. Squall. A squall is a strong wind characterized by a sudden onset in wind speed increasing at least 16 knots and sustained at 22 knots or more for at least 1 minute.

A2.1.5. Volcanic Eruption. Eruption or volcanic ash cloud first noted.

A2.1.6. Thunderstorm.

A2.1.6.1. Begins (**Note:** A SPECI is not required to report the beginning of a new thunderstorm if one is currently reported as in progress at the airfield).

A2.1.6.2. Ends (**Note:** 15 minutes after the last occurrence of criteria for a thunderstorm; an audible sound of thunder, lightning within five (5) NM of the airfield etc.).

A2.1.7. Precipitation.

A2.1.7.1. Hail begins or ends.

A2.1.7.2. Freezing precipitation begins, ends, or changes intensity.

A2.1.7.3. Ice pellets begin, end, or change in intensity.

A2.1.7.4. Any other type of precipitation begins or ends. Note: Except for freezing rain, freezing drizzle, hail, and ice pellets, a SPECI is not required for changes in type (e.g., drizzle changing to snow grains) or the beginning or ending of one type while another is in progress (e.g., snow changing to rain and snow).

A2.1.8. Tornado, Funnel Cloud, or Waterspout. When a tornado is first observed, disappears from sight, or ends.

A2.1.9. Runway Visual Range (RVR). WF provides RVR output according to the specifications listed in **Table A2.3**.

Table A2.3. RVR Reporting, Part 1.

Runway Visual Range (RVR)	
- RVR for active runway decrease to less than or, if below, increase to equal or exceed:	
- 6,000 feet (P1500 meters)	- 2,000 feet (0600 meters)
- 5,500 feet (P1500 meters)	**- 1,800 feet (0550 meters)**
- 5,000 feet (1500 meters)	- 1,600 feet (400 meters)
- 4,500 feet (1400 meters)	**- 1,200 feet (0350 meters)**
- 4,000 feet (1200 meters)	- 1,000 feet (320 meters)
- 2,400 feet (0750 meters)	- 600 feet (200 meters)
- Items in bold indicate criteria found in the high and low altitude FLIPs. - All published RVR minima applicable to the runway in use.	

Table A2.4. RVR Reporting, Part 2.

Runway Visual Range (RVR)	
- RVR is first determined as unavailable (RVRNO) for the runway in use, and when it is first determined that the RVRNO report is no longer applicable, provided conditions for reporting RVR exist.	

A2.1.10. Tower Visibility. When notified by ATC that the tower prevailing visibility has decreased to less than or, if below, increased to equal or exceed 1, 2, 3, or 4 statute miles and the tower prevailing visibility differs from the surface prevailing visibility.

A2.1.11. Upon Resumption of Observing Services. Take, disseminate, and record a SPECI within 15 minutes after returning to duty following a break in hourly coverage, if a METAR was not filed as scheduled during the 15-minute period.

A2.1.12. Aircraft Mishap. When notified of an aircraft mishap, the WF checks the latest AN/FMQ-19 observation (i.e., METAR/SPECI/OMO (one minute observation)) and performs

augmentation/back-up if required. When operating in a back-up mode WF immediately takes a SPECI observation IAW **AFMAN 15-111**.

A2.1.13. Any other meteorological situation that, in the weather technician's opinion, is critical.

Attachment 3

FORECAST SPECIFICATION AND AMENDMENT CRITERIA

A3.1. TAFs specify the onset, duration, and intensity for the criteria listed below. TAFs are amended when forecast/observed conditions do not match conditions specified in the TAF.

A3.1.1. **Ceiling and/or Visibility.** Forecast to decrease to less than; if below, is forecast to increase to equal or exceed the thresholds listed in **Table A3.1**.

Table A3.1. Ceiling and Visibility Specification and Amendment Criteria.

Ceiling	Visibility	Category	Reference
Ceiling and/or visibility observed or expected to decrease to less than, or if below, increase to equal or exceed.			
≥ 2,000 FT	≥ 3 SM (4,800 M)	E	AFMAN 15-129V1
< 2,000 FT but ≥ 1,000 FT	< 3 SM (4,800 M) but ≥ 2 SM (3200 M)	D	AFMAN 15-129V1
< 1,000 FT but ≥ 700 FT	< 3 SM (4,800 M) but ≥ 2 SM (3200 M)	C	AFMAN 15-129V1
< 700 FT but ≥ 200 Ft	< 2 SM (3,200 M) but ≥ 1/2 SM (0800 M)	B	AFMAN 15-129V1
< 200 FT	< 1/2 SM (800 M)	A	AFMAN 15-129V1

A3.1.2. Surface Winds.

A3.1.2.1. Wind Speed: The difference between the predominant speed and the forecast speed is ≥ 10 knots.

A3.1.2.2. Wind Gusts: The difference between the observed gusts and the forecast gusts is ≥ 10 knots.

A3.1.2.3. Wind Direction: A change in the predominant direction of > 30 degrees when the wind speed or gusts are forecast to be ≥ 15 knots.

A3.1.3. Icing. Not associated with thunderstorms, from the surface to 10,000 ft Above Ground Level (AGL).

A3.1.3.1. When the beginning of icing first meets or exceeds the moderate or greater threshold and was not specified in the forecast.

A3.1.3.2. When icing decreases to less than the moderate threshold and was not specified in the forecast.

A3.1.4. Turbulence. For turbulence category II aircraft, not associated with thunderstorms, from the surface to 10,000ft AGL.

A3.1.4.1. When the beginning of turbulence first meets or exceeds the moderate or greater threshold and was not specified in the forecast.

A3.1.4.2. When turbulence decreases to less than the moderate threshold and was not specified in the forecast.

A3.1.5. Weather Warning and Forecast Weather Advisory Criteria.

A3.1.5.1. Occur, or are expected to occur during the forecast period, but were not specified in the forecast.

A3.1.5.2. Specified in the forecast but are no longer expected to occur during the forecast period.

A3.1.6. Altimeter setting.

A3.1.6.1. If below, increases to meet or exceed 31.00 INS and was not specified during the forecast period.

A3.1.6.2. If above, decreases to less than 31.00 INS and was not specified during the forecast period.

A3.1.6.3. If above, decreases to less than 28.00 INS and was not specified during the forecast period.

A3.1.6.4. If below, increases to equal or exceed 28.00 INS and was not specified during the forecast period.

A3.1.7. Thunderstorms.

A3.1.7.1. If the start time is incorrectly forecast.

A3.1.7.2. If the end time was incorrectly forecast.

A3.1.8. Temporary Conditions.

A3.1.8.1. Conditions specified as temporary become predominant.

A3.1.8.2. Conditions specified as temporary do not occur during the cardinal hour as forecast.

A3.1.8.3. Conditions specified as temporary are no longer expected to occur.

A3.1.9. Predominant Conditions.

A3.1.9.1. Change in conditions occurs before the beginning of the specified period of the change and are expected to persist.

A3.1.9.2. Change in conditions does not occur within 30 minutes after the specified time.

A3.1.9.3. Change in conditions are no longer expected to occur.

A3.1.10. Representative Conditions. Conditions are not considered representative of existing or forecast conditions and amending the forecast improves safety, flight planning, operations efficiency, or assistance to an in-flight aircraft.

Attachment 4

SAMPLE WEATHER PRODUCT DISSEMINATION FORMAT/INTERPRETATION OBSERVATION/TAF/WWAS- SPECI KSUU 1506Z AUTO 17013G22KT 2 1/2 RVRNO TSRA BKN015CB OVC030 19/18 ALSTG 29.99 RMK AO2 TS OHD MOV NE

Table A4.1. Sample Weather Observations.

Body of Report - Consists of 11 Groups		
Group	Reference	Brief Description
Type of Report	A4.1.	Indicates type of report.
Station Identifier	A4.2.	A four-character group used to identify the observing location.
Date and Time of Report	A4.3.	Date and time of the report.
Report Modifier	A4.4.	A report modifier (**COR**) identifying report as a correction, or **AUTO** indicating the weather observation is a fully automated report with no human intervention. Gusts are appended if available.
Wind	A4.5.	Indicates wind direction and speed.
Visibility	A4.6.	Provides prevailing visibility from the designated point of observation in statute miles or meters.
Runway Visual Range	A4.7.	10-minute RVR or varying RVR in hundreds of feet or meters.
Present Weather	A4.8.	Any weather occurring at the observing location, obscurations to vision, or other phenomena.
Sky Condition	A4.9.	State of the sky in terms of sky cover, layers and heights, ceilings and obscurations.
Temperature and Dew Point	A4.10.	Measure of hotness/coldness of ambient air. Dew point measures saturation point temperature.
Altimeter	A4.11.	Indicates altitude above MSL of an aircraft on the ground.
Remarks	A4.12.	Remarks generally elaborate on parameters reported in the body of the report, and are included in all METAR and SPECI observations.

A4.1. Type of Report. METAR or SPECI.

A4.2. Station identifier, also called the ICAO. This code identifies the location of the observation.

A4.3. Date and Time of Report. This is in Zulu (GMT) of the last element of the observation.

A4.4. Report Modifier. The report modifier can be either of the following two elements:

A4.4.1. COR is entered into the report modifier group when a corrected METAR or SPECI is transmitted.

A4.4.2. AUTO identifies the report as a fully automated report with no human intervention. It automatically included in reports when the weather technician signs off the AMOS indicating the observations are no longer being augmented.

A4.5. Wind. The true direction the wind is blowing from is encoded in tens of degrees using three Tables. Directions less than 100 degrees are preceded with a "0." The wind speed is entered as a two or three-digit group immediately following the wind direction.

A4.5.1. Gust. The wind gust is encoded in two or three digits immediately following the wind speed. The wind gust is encoded in whole knots using the units and tens digits and, if required, the hundreds digit.

A4.5.2. Variable Wind Direction (speeds 6 knots or less). Variable wind direction with wind speed 6 knots or less may be encoded as VRB in place of the direction.

A4.5.3. Variable Wind Direction (speeds greater than 6 knots). Wind direction varying 60 degrees or more with wind speed greater than 6 knots are encoded as variable. The variable wind direction group is immediately following the wind group. The directional variability is encoded in a clockwise direction. For example, if the wind is variable from 180 degrees to 240 degrees at 10 knots, it would be encoded 21010KT 180V240.

A4.5.4. Calm Wind. Calm wind is encoded as 00000KT.

A4.6. Visibility. The furthest predominant distance (at least 50% of the aerodrome) seen from the airfield reported in statute miles. The most common visibility reported is 10 miles.

A4.7. Runway Visual Range.

A4.8. Present weather. Any weather phenomenon occurring on the airfield. This is mandatory anytime the visibility is less than 7 miles. **Table A4.2** lists the present weather codes:

Table A4.2. Weather Phenomena Codes.

Qualifier	Weather Phenomena			
Intensity Or Proximity	**Descriptor**	**Precipitation**	**Obscuration**	**Other**
- Light	MI (Shallow)	DZ (Drizzle)	BR (Mist)	PO (Developed Dust/Sand Whirls)
Moderate	PR (Partial)	RA (Rain)	FG (Fog)	SQ (Squall)
+ Heavy	BC (Patches)	SN (Snow)	FU (Smoke)	FC (Funnel Cloud, Tornado, or Water Spout)
VC (Vicinity)	DR (Low Drifting)	SG (Snow Grains)	VA (Volcanic Ash)	SS (Sandstorm)
	BL (Blowing)	IC (Ice Crystals)	DU (Dust)	DS (Dust Storm)
	SH (Showers)	PL (Ice Pellets)	SA (Sand)	
	TS (Thunderstorms)	GR (Hail)	HZ (Haze)	
	FZ (Freezing)	GS (Small Hail or Snow Pellets)	PY (Spray)	
		UP (Unknown Precip)		

A4.9. Sky Condition and Cloud Height. Describes the amount of clouds present at the airfield and the base of each cloud deck. They fall into the following categories:

A4.9.1. CLR – Sky Clear.

A4.9.2. FEW – 1/8 to 2/8 coverage.

A4.9.3. SCT – Scattered; 3/8 to 4/8 coverage.

A4.9.4. BKN – Broken; 5/8 to 7/8 coverage.

A4.9.5. OVC – Overcast; 8/8 coverage.

A4.9.6. VV – Vertical visibility; normally used during heavy fog, indicates how far up into the fog can be seen.

A4.9.7. FEW000, SCT000, BKN000 – Surface-based obscuration.

A4.9.8. Cloud Height. Three-digit number provides the height of the base of the cloud in hundreds of feet (e.g., 015 equals 1,500 feet). The CB and TCU descriptors may be appended to the cloud height to indicate the cloud is a cumulonimbus or towering cumulus.

A4.10. Temperature and Dew Point. (i.e., can be in degrees, either Fahrenheit or Celsius).

A4.11. Altimeter Setting. The current value aircraft altimeters must be set at to read an elevation of zero. The altimeter is measured in inches (INS) of mercury.

A4.12. Remarks. Table A4.3 contains some of the most commonly seen remarks in observations:

Table A4.3. Remarks Listing.

AO2—Automated sensor indicator
CB—Cumulonimbus
DSNT—Distant
ESTMD—Estimated
FROPA—Frontal Passage
LTG—Lightning
LWR—Lower
MOV—Moving
MOVD—Moved
OHD—Overhead
PK WND—Peak Wind
PRESFR—Pressure Falling Rapidly
PRESRR—Pressure Rapidly Rising
RWY—Runway
TCU—Towering Cumulus
TWR—Tower
UNKN—Unknown
VIS—Visibility
WSHFT—Wind Shift
PA—Pressure Altitude
DA—Density Altitude

A4.13. TAF.

Table A4.4. Sample TAF.

TAF KSUU 0313/0419 14006KT 9999 BKN040 OVC100
QNH2996INS BECMG 0318/0319 20010G15KT 9999 BKN040
620406 QNH2993INS
BECMG 0320/0321 19015G20KT 8000 -SHRA FEW025 OVC040 610406
QNH2991INS TEMPO 0323/0403 VRB10G20KT 3200 -TSRA BKN015CB BKN035
BECMG 0404/0405 21007KT 9000 -RA FEW025 BKN060 620604 QNH2992INS
BECMG 0417/0418 21012KT 9999 NSW FEW030 BKN050 620505
QNH2997INS TX12/0322Z TN03/0313Z

A4.13.1. The forecast follows the same general format as the observation with the following exceptions noted:

A4.13.1.1. Valid Date/Time. Forecasts are valid for a 30-hour period. In this example, the forecast is valid from the third at 1300Z until the fourth at 1900Z.

A4.13.1.2. BECMG – This is a code to indicate the predominant conditions are going to change to (or become) the conditions listed in the line of the forecast. The conditions are changing during the time period following the BECMG code (1800 to 1900Z in the example above).

A4.13.1.3. TEMPO – This code means the conditions listed on the line may occur for periods of less than 30 consecutive minutes (45 minutes or less for thunderstorms) anytime between the time frame following the TEMPO code (03/2300Z to 04/0300Z in this example).

A4.13.1.4. Max Temp/Min Temp. TX12 indicates a maximum temperature of 12° Celsius to occur at 22Z. TN03 indicates a minimum temperature of 03° Celsius to occur at 13Z (Note: "M" indicates a minus sign in front of the number. Example: M05 = -5° Celsius).

A4.14. Weather Warnings, Watches, and Advisories.

Table A4.5. Observed Weather Warning.

TRAVIS AFB WEATHER WARNING 05-001
VALID 17/1921Z (17/1321L) TO UFN LIGHTNING IS OBSERVED WITHIN 5NM
08/RS
1. FORECAST WEATHER WARNING.
TRAVIS AFB WEATHER WARNING 11-051
VALID 10/1500Z(10/0900L) TO 10/2200Z(10/1600L)
WINDS ASSOCIATED WITH MODERATE THUNDERSTORMS ARE FORECAST TO BE 35-49 KNOTS AT TRAVIS AFB. MAXIMUM GUST EXPECTED: 41 KNOTS
18/JDB

2. WEATHER WATCH.
TRAVIS AFB WEATHER WATCH 05-215
VALID 15/1858Z (15/1358L) TO 15/2100Z (15/1600L)
A LIGHTNING WATCH IS NOW IN EFFECT FOR TRAVIS AFB UNTIL 1600L.
A WARNING WILL BE ISSUED LATER IF REQUIRED.
58/AW

3. OBSERVED WEATHER ADVISORY.
TRAVIS AFB WEATHER ADVISORY 09-134
VALID 08/1408Z (080908L) TO UFN
CROSSWINDS OBSERVED TO BE EQUAL TO OR GREATER THAN 27KTS
44/JP

Attachment 5

CUSTOMER RESPONSE MATRIX

Table A5.1. Customer Response Matrix, Part 1.

Weather Phenomena	Lead Time	Impact	Customer Action
Tornado	15 min	Personal injury/death Equipment damage	Seek shelter
Damaging Hail (≥ 3/4")	1 hr	Personal injury/death Equipment damage	Seek shelter; hangar or divert aircraft
Freezing Precipitation	1 hr	Delay or cease operations	Cease flying; hangar or protect aircraft
Damaging Winds ≥ 50 knots	2 hrs	Flight hazard Equipment damage	Cease unnecessary flying; secure or hangar aircraft; secure light objects outside
Strong Winds 35-49 knots	1 ½ hrs	Flight hazard Equipment damage	Cease unnecessary flying; secure or hangar aircraft
Surface winds 25-34 knots	30 min	Delay operations	Aircraft Mx: Do not operate high reach equipment. Remove all maintenance equipment not in use from aircraft parking area. Do not change positions of cowlings. Down-jack aircraft ASAP. Retract slats/flaps. Close aircraft hangar doors.
Lightning w/in 5 NM of Travis AFB	Observed	Personal injury/death Delay operations	Cease flight-line work; clear pool/golf course
Moderate Icing below 10,000 ft AGL	Observed	Flight hazard	Cease unnecessary flying. Evaluate icing conditions.
Moderate Turbulence below 10,000 ft AGL	Observed	Flight hazard	Cease unnecessary flying. Evaluate turbulence conditions.
* Crosswinds ≥ 31 knots	Observed	Flight hazard	Cease/delay take-off for KC-10, C-17, and E-6B aircraft
Crosswinds ≥ 27 knots	Observed	Flight hazard	Cease/delay take-off for C-5 aircraft

Table A5.2. Customer Response Matrix, Part 2.

Weather Phenomena	Lead Time	Impact	Customer Action
Crosswinds ≥ 19 knots but < 27 knots **(wet runway)**	Observed	Flight hazard	Cease take-off/landings for KC-10, and C-5 aircraft
Crosswinds ≥ 12 knots but < 19 knots **(wet runway)**	Observed	Flight hazard	Cease take-off/landings for Navy E-6B aircraft
Low Level Wind Shear	Observed	Delay or cease operations	Delay or cease take-off/landing evaluate shear conditions

A5.1. Note: The advisory for observed crosswinds ≥ 31 knots is specifically for KC-10 and C-17 aircraft; however, the E-6B threshold is ≥ 32 knots—an additional advisory is unnecessary.

Attachment 6

FLYING UNITS SUPPORTED & MISSION LIMITING ENVIRONMENTAL CONDITIONS

Table A6.1. Flying Units Supported.

Organization	Mission	MWP Provider
21 AS and 301 AS (C-17)	Provides worldwide airlift for war, peacetime, and contingency operations.	Travis AFB WF/TACC/XOW
22 AS and 312 AS (C-5)	Provides worldwide airlift for war, peacetime, and contingency operations.	Travis AFB WF/TACC/XOW
6, 9, 70, and 79 ARS (KC-10)	Provides worldwide airlift for war, peacetime, and contingency operations.	Travis AFB WF/TACC/XOW
Navy VQ-3 (E-6B)	Provides C2 for POTUS	Navy

A6.1. Mission Limiting Thresholds.

A6.1.1. Airframe-Specific Weather Limitations. **Table A6.2** and **Table A6.1** provide the general airframe weather limitations based on **AFMAN 11-202V3**, *Flight Operations.*

Table A6.2. USAF General Flight Rules Weather Limitations.

(Ref: AFI 11-202V3)		
Weather Condition	Impact	Customer Action
Cig/Vis < 2,000 / 3	Alternate required	Add fuel to allow divert
Cig/Vis < 1,000/ 2, if MAJCOM approved	Alternate required	Add fuel to allow divert
Cig/Vis < 500 / 2	Terminal not suitable for alternate	Select another alternate

Table A6.3. Travis AFB Airframe GO/NO-GO Thresholds.

	KC-10 (Cat III)	C-5 (Cat II)	C-17 (Cat III)	E-6B (Cat II)
TAKEOFF AND LANDING THRESHOLDS	CIG <200FT / VIS <1/2SM			
	LOW-LEVEL WIND SHEAR			
	THUNDERSTORMS			
	SEVERE TURBULENCE			
	SEVERE ICING			
	WR// 19KT X-WIND	WR// 19KT X-WIND	X-WIND 31KT (WR// OR DRY)	WR// 12KT X-WIND
	X-WIND 31KT	X-WIND 27KT		X-WIND 32KT
ENROUTE HAZARDS	SEVERE TURBULENCE			
	SEVERE ICING			
	THUNDERSTORMS			
	SCT TS	SCT TS	SCT TS	AVOID ALL TS
AR HAZARDS	TS W/IN 10NM OF FL <230, 20NM ABOVE FL230	TS W/IN 10NM OF FL <230, 20NM ABOVE FL230	TS W/IN 10NM OF FL <230, 20NM ABOVE FL230	AVOID ALL TS
	AR VIS <1NM			
	MODERATE ICING IN AR			
	MODERATE TURBULENCE IN AR (OBSERVED)			
	SEVERE TURBULENCE IN AR (FORECAST)			
VR/SR ROUTES (NO CIG/VIS LIMS FOR IR ROUTES)			CIG/VIS LTE 1500FT/3SM	
			AVOID TS BY 5NM IF OUTSIDE AIR TEMP IS ABOVE 0C AT FL. IF NOT, AVOID ALL TS	
NOTE 1: RVR LIMITS ARE FOR GENERAL INFORMATION ONLY. WF PERSONNEL DO NOT ISSUE RVR FORECAST OR MISSIONWATCH FOR RVR MINIMA				
RVR LIMITS (NO RVR RWY 21R)	21L/03R: 1000 TOUCHDOWN, 1000 ROLLOUT	21L/03R: 1000 TOUCHDOWN, 1000 ROLLOUT	21L/03R: 1200 TOUCHDOWN, 1000 ROLLOUT	21L/03R: 1000 TOUCHDOWN, 1000 ROLLOUT
	03L: 1600			

Attachment 7

MISSION EXECUTION FORECAST EXAMPLE

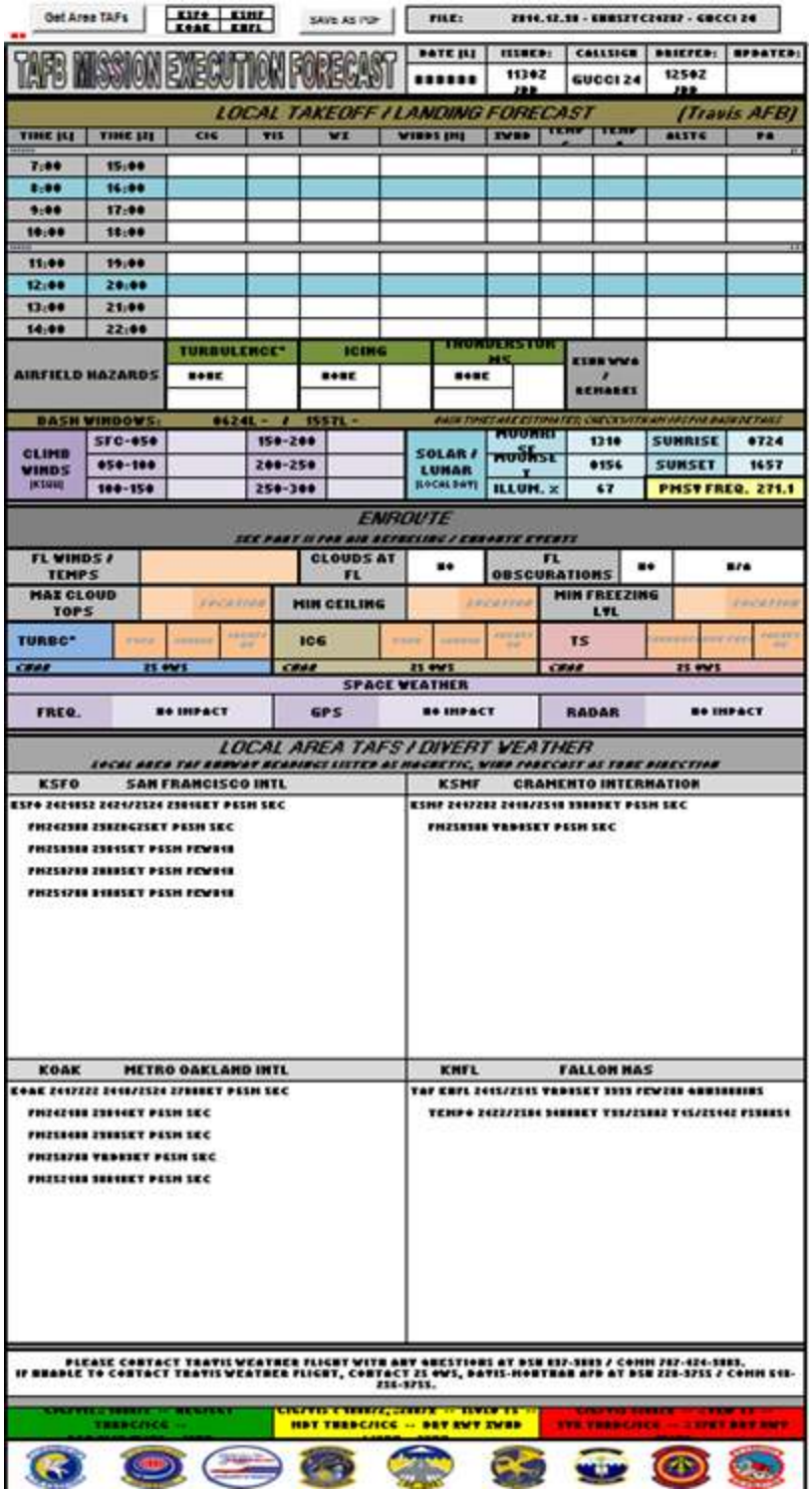
Table A7.1. Travis AFB MWP.



Attachment 8

MISSION EXECUTION FORECAST BACK UP EXAMPLE

Table A8.1. Travis AFB MEF Backup Example.



Get Area TAFs | KSFO KSMF KOAK KMFL | Save As PDF | FILE: 2016.12.30 - KBRSZTC24282 - GUCCI 24

TAFB MISSION EXECUTION FORECAST

DATE (L)	ISSUED:	CALLSIGN	BRIEFED:	UPDATED:
888888	1130Z JBB	GUCCI 24	1250Z JBB	

LOCAL TAKEOFF / LANDING FORECAST (Travis AFB)

TIME (L)	TIME (Z)	CIG	VIS	WX	WINDS (M)	ZWND	TEMP F	TEMP C	ALSTG	PA
7:00	15:00									
8:00	16:00									
9:00	17:00									
10:00	18:00									
11:00	19:00									
12:00	20:00									
13:00	21:00									
14:00	22:00									

AIRFIELD HAZARDS	TURBULENCE*	ICING	THUNDERSTORMS	ESBR VVA / REMARKS
	###E	###E	###E	

BASH WINDOWS: 0624L - / 1557L -

CLIMB WINDS (KTAS)					SOLAR / LUNAR (LOCAL DAY)				
	STC-050		150-200		MOONRISE	1210	SUNRISE	0724
	050-100		200-250		MOONSET	0156	SUNSET	1657
	100-150		250-300		ILLUM. %	67	PMSV FREQ. 271.1	

ENROUTE

SEE PART II FOR AIR REFUELING / CURRENT EVENTS

FL WINDS / TEMPS		CLOUDS AT FL	##	FL OBSCURATIONS	##	#/#
MAX CLOUD TOPS	LOCATION	MIN CEILING	LOCATION	MIN FREEZING LVL	LOCATION	
TURBC*	TIME	CONDS	FREQ OF OCCUR	ICG	TIME	CONDS	FREQ OF OCCUR	TS	CONDS	FREQ OF OCCUR
CMNT	25 OWS	CMNT	25 OWS	CMNT	25 OWS					

SPACE WEATHER

FREQ.	## IMPACT	GPS	## IMPACT	RADAR	## IMPACT

LOCAL AREA TAFS / DIVERT WEATHER

LOCAL AREA TAF RUNWAY HEADINGS LISTED AS MAGNETIC, WIND FORECAST AS TRUE DIRECTION

KSFO SAN FRANCISCO INTL	KSMF CRAMENTO INTERNATION
KSFO 2621052 2621/2524 23010KT P6SM SKC	KSMF 2617202 2618/2518 33003KT P6SM SKC
FM242300 25020G25KT P6SM SKC	FM253000 VRB05KT P6SM SKC
FM253000 23015KT P6SM FEW010	
FM259700 28005KT P6SM FEW010	
FM251700 01005KT P6SM FEW010	

KOAK METRO OAKLAND INTL	KMFL FALLON NAS
KOAK 2617222 2618/2524 27008KT P6SM SKC	TAF KMFL 2615/2515 VRB05KT 9999 FEW200 ########
FM242100 23014KT P6SM SKC	TEMPO 2622/2504 340##KT 7SM/25002 Y15/25102 P1000SM
FM254000 25005KT P6SM SKC	
FM258700 VRB05KT P6SM SKC	
FM252100 30010KT P6SM SKC	

PLEASE CONTACT TRAVIS WEATHER FLIGHT WITH ANY QUESTIONS AT DSN 837-5883 / COMM 707-424-5883.
IF UNABLE TO CONTACT TRAVIS WEATHER FLIGHT, CONTACT 25 OWS, DAVIS-MONTHAN AFB AT DSN 228-5755 / COMM 520-228-5755.

TSTM/ICG -- | HVY TS / SMALL / LARGE HAIL -- HVY TS -- HVY RWY ICING | SVR TSTM/ICG -- 2 OR MORE RWY

Attachment 9

TRAVIS AFB METEOROLOGICAL EQUIPMENT LOCATIONS

Figure A9.1. Travis AFB Airfield FMQ-19 Sensor Groups.

